

Full Length Article

Dilated multi-Layer perceptron mixer for faster neural networks

Van-Dung Hoang ^{a,*}, Xuan-Thuy Vo ^b, Kang-Hyun Jo ^b^a Faculty of Information Technology, Ho Chi Minh City University of Technology and Engineering, Ho Chi Minh City, Vietnam^b Department of Electrical Electronic and Computer Engineering, University of Ulsan, Ulsan, South Korea

ARTICLE INFO

Keywords:

Deep learning
Image classification
Instance segmentation
MLP-Mixer
Object detection

ABSTRACT

Multi-Layer Perceptron Mixer (MLP-Mixer) has replaced self-attention layers with simple spatial MLP and achieved promising performances across visual tasks. However, existing MLP-Mixer methods have quadratic complexity with image resolutions and only process the input with a fixed size. When transferring pre-trained MLP-Mixer to downstream tasks, the model creates high computational costs and requires further operations to interpolate the MLP weights. This paper addresses these issues by proposing Dilated MLP-Mixers (DMLP) method, faster backbones for image classification and downstream tasks. DMLP backbone performs spatial mixing on non-overlapped and dilated windows to achieve better efficiency while exchanging information across windows. Unlike the MLP-based traditional approaches such as the Spatial MLP-Mixer architecture, the proposed method creates sparse and cross-connections between windows in the window MLP-Mixer block, known as the dilated windows. This approach allows for reducing the number of windows while it does not increase the number of connections within windows compared to previous approaches. This design has flexibility to architecture at different input sizes and results in linear complexity with image resolutions. The proposed method has been validated in the domain of automobile insurance and car damage recognition. Extensive experiments were conducted on benchmark datasets to evaluate its performance. Results indicate that the proposed DMLP backbone achieves performance comparable to MLP and Transformer models while demonstrating superior processing speed. Comparative evaluations reveal that the proposed method outperforms recent approaches in terms of both accuracy and computational efficiency. Furthermore, this method shows potential for application in systems with limited computational resources.

1. Introduction

The problem of classifying images plays a crucial role in the field of computer vision. It is one of the important applications of artificial intelligence (AI) and image processing. The problem involves advanced feature analysis techniques to increase the accuracy of pattern recognition. In recent years, deep learning (DL) techniques based on convolutional neural networks (CNN), Vision Transformers (ViT) have brought significant advancements. CNNs He et al. (2016), Sandler et al. (2018) dominated visual models in feature extractors. Deep learning models based on CNNs have the capability to autonomously learn hierarchical features from image data by utilizing convolutional layers, achieving high accuracy. They have numerous applications such as disease diagnosis in healthcare, product monitoring, fault detection, and anomaly detection in objects such as Peng et al. (2023), Pham and Hoang (2024), Xu et al. (2024). In the application of CNNs, the authors in Xu et al. (2024) combine convolutional neural networks with an online soft-label training strategy for mechanical fault diagnosis. This method aims to achieve

modality-specific and cross-modal feature representation using multi-source data. It integrates features from both independent and shared mechanical signals. Experimental results on the NU205E and HF-MS Dataset demonstrate superior performance compared to state-of-the-art methods, particularly in noisy conditions. Meanwhile, the authors Peng et al. (2023) utilized a technique combining semi-supervised learning for the emotion recognition task. The proposed model integrates EEG feature transfer and semi-supervised learning into a single objective for improving overall performance by avoiding suboptimality caused by sequential. The experimental results on benchmark datasets demonstrated that the proposed method is effective and outperforms in the field of using EEG features for emotion recognition.

In recent years, Vision Transformer (ViT) Dosovitskiy et al. (2021) has successfully adopted Transformer Yao et al. (2024) into vision tasks. Additionally, MLP-Mixer techniques have also shown effectiveness in image feature extraction tasks despite not relying on CNN or Self-attention principles. The MLP-Mixer Tolstikhin et al. (2021), Zhuang et al. (2024) architecture is considered as an alternative to complex

* Correspondence author

E-mail address: dunghv@hcmute.edu.vn (V.-D. Hoang).<https://doi.org/10.1016/j.neunet.2026.108939>

Received 10 July 2025; Received in revised form 27 March 2026; Accepted 2 April 2026

Available online 3 April 2026

0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1
Comparison of Spatial MLP-Mixer and DMLP-Mixer modules.

Module	Local	Global	Adaptive	Complexity $\mathcal{O}$
Spatial MLP-Mixer	✗	✓	✗	N^2
Swin-MLP Liu et al. (2021b)	✓	✓	✗	N
AS-MLP Lian et al. (2022)	✓	✗	✓	N
CycleMLP Chen et al. (2023)	✓	✓	✓	N
Hire-MLP Guo et al. (2022)	✗	✓	✓	N
DynaMixer Wang et al. (2022b)	✗	✓	✗	N
RaMLP Lai et al. (2023)	✓	✓	✓	N^2/d_r
DMLP-Mixer	✓	✓	✓	N/d_r

architectures such as CNN or Self-attention. Regarding the technical aspect of transfer learning, authors in Yao et al. (2024) present the method based on prior normality prompt transformer for multiclass image anomaly detection. It incorporates normal semantics prompting to mitigate the identical mapping problem by integrating prior normality prompts into the reconstruction process, creating a dual-stream model. Experimental results demonstrated that it is effective across various benchmark datasets and real-world scenarios, positioning it as a state-of-the-art solution in the field. Authors in Zhang et al. (2024) adopt an approach using MLP-Mixer for the Visual Fault Detection of Freight Trains. The MLP neural network was specifically designed with spatial-wise dynamic distillation. This approach extracts richer details from lower-level feature appearances and incorporates higher-level label semantics as additional supervision signals. It utilizes efficient instance embedding to model global spatial and semantic information. The method outperforms the state-of-the-art approach for detectors and achieves high accuracy while enabling real-time detection at a reduced computational cost. In contrast, authors in Jiang et al. (2023) apply the Masked Swin Transformer Unet approach for the Industrial Anomaly Detection problem. The model is trained on a particularly insufficient abnormal dataset. The primary approach leverages Swin Transformer to achieve high accuracy, combined with the Unet network to construct an end-to-end anomaly detection solution. Experimental results on the MVTec AD dataset demonstrate that the method achieves high performance in both anomaly detection and localization.

However, it has proven to be effective in extracting relationships between data elements in images and achieving high accuracy for classification tasks. The architecture of Spatial MLP-Mixer relies on multi-layer perceptron networks (MLPs), and these blocks are repeatedly applied across spatial locations or feature channels. Although Spatial MLP-Mixer attains promising performances on image classification tasks, Spatial MLP-Mixer has two main drawbacks:

- Firstly, the learnable matrix is fixed during training and testing. To align the size between two matrices, the input sizes are matched with the weight dimension. When transferring Spatial MLP-Mixer to dense prediction tasks, e.g., object detection and segmentation, the models require further operations to interpolate weight. This reduces the performance and also adaptation ability for resource-constrained systems.
- Secondly, matrix multiplication between learnable matrix and input results in quadratic complexity and high number of parameters. Transferring Spatial MLP-Mixer to dense prediction produces extremely high computation costs and requires large GPU memories to train the model.

One solution based on MLP-Mixer is to analyze local relationships within smaller-sized windows to reduce computational complexity. However, this approach sacrifices global relationships between image regions and remains high complexity.

To address this problem, a novel approach has been developed in a new research direction by proposing Dilated MLP-Mixers (DMLP). The DMLP network is a hierarchical architecture consisting of four stages, where convolutional blocks are used in stages 1 and 2 for local feature extraction on high-resolution inputs, while Dilated MLP blocks

are stacked in stages 3 and 4 to capture global features with linear complexity. In stages 3 and 4, the DMLP block performs spatial mixing within non-overlapping windows and dilates windows to achieve better efficiency while exchanging cross-window information. This approach differs from traditional MLP-based methods such as the Spatial MLP-Mixer architecture; the proposed method creates sparse and cross connections between windows in the window MLP-Mixer block. This solution is implemented based on the principle of cross-feature fusion between windows to increase efficiency in information extraction and expand links to accelerate computation. Thus, this approach reduces the total number of windows needed by using dilation, while maintaining the same intra-window connection density, resulting in an overall complexity reduction from $\mathcal{O}(N)$ to $\mathcal{O}(N/d_r)$ (d_r is the dilation rate). This design type provides flexibility in architecture with different input sizes and linear complexity with input image resolution. Recent MLP methods have explored different strategies for handling variable input sizes: AS-MLP Lian et al. (2022) uses axial shifts (but limited to local dependencies), CycleMLP Chen et al. (2023) employs cyclic sampling (which may lose small object cues), and RaMLP Lai et al. (2023) proposes dilated fully-connection (achieving $\mathcal{O}(N^2/d_r)$ complexity). While these methods address input flexibility, they either sacrifice global modeling or maintain quadratic complexity dependencies. DMLP uniquely achieves $\mathcal{O}(N/d_r)$ strictly linear complexity by dilating window partitioning rather than feature partitioning or connectivity matrices, offering superior efficiency for resource-constrained applications. Table 1 shows comparison between our DMLP-Mixer module and other MLP modules where N denotes number of tokens, d indicates channel dimension, and k is the window size. DMLP-Mixer captures both local and global interactions at lower computational cost and number of parameters compared to Spatial MLP-Mixer. For example, at stage 3 of the classification network, the input feature has the spatial size of 28×28 and 96 channels, and the window size is configured to 7. Spatial MLP-Mixer creates 59.0 MFLOPs and 614.6K parameters while DMLP-Mixer module only produces 3.7 MFLOPs and 2.4K parameters. It verifies that the proposed DMLP-Mixer achieves high efficiency while capturing more spatial interactions between tokens. We have described the main contributions of our paper as follows:

1. This paper proposes a new MLP module, dubbed Dilated MLP-Mixer (DMLP). While window-based partitioning has been explored in Swin-MLP, our core contribution lies in introducing a dilated window sampling mechanism specifically designed for MLP-based spatial mixing. Unlike Swin-MLP shifted window approach with masked attention, DMLP employs dilated window sampling that reduces the number of windows from N_w to N_w/d_r , enabling efficient cross-window information exchange without attention mechanisms. This design reduces the quadratic complexity of Spatial MLP-Mixer from $\mathcal{O}(N^2)$ to $\mathcal{O}(N/d_r)$ while maintaining both local and global modeling capabilities.
2. Based on the proposed MLP module, we introduce an efficient and general-purpose DMLP-Mixer backbone. DMLP-Mixer is a hierarchical backbone consisting of convolutional blocks at earlier stages and DMLP blocks at later stages while MLP-mixer variants only extract features on single-scale input. DMLP-mixer backbone processes the input with various sizes and eliminates the need for interpolation operations during transfer learning. Recent MLP-Mixers require interpolation operation to resize the weights of models when dealing with different input sizes.
3. Extensive experiments are validated across general-purpose datasets (ImageNet-1K, MS-COCO, ADE-20K) and specialized applications (car damage recognition), showing consistent improvements in both accuracy and efficiency. Compared to CNN-based and Transformer-based backbones, DMLP-Mixer achieves similar or even better accuracy while getting higher efficiency.

2. Related works

This work proposes an efficient feature extractor based on MLP-Mixer. There are three main methods of feature extractors: convolution, self-attention, and MLP. Hence, we organize Related Works section into three categories: Convolutional Neural Networks, Transformer-based approaches, and Spatial MLP-Mixer.

2.1. Convolutional neural networks

In the field of pattern classification and recognition using deep neural networks (DNN), CNN models have been proposed to achieve high performance as in He et al. (2016), Howard et al. (2017), Hu et al. (2024), Simonyan and Zisserman (2014), Zhou et al. (2024). VGGNet Simonyan and Zisserman (2014) proposed plain networks with the combination of convolution, batch normalization, and ReLU activation function. ResNet He et al. (2016) observed gradient vanishing when stacking more layers in VGG and proposed a simple residual connection to overcome this issue. With this design, ResNet became the common backbone for many vision tasks Pham et al. (2021), Xu et al. (2024). ResNet incorporates short connections within the neural network, thus alleviating the vanishing gradient problem when constructing deeper network architectures. MobileNet Howard et al. (2017) consists of depth-wise separable convolutional layers as an effective replacement for traditional convolutional layers. UIU-Net Wu et al. (2022) improves the multi-scale feature extraction of U-Net architecture by incorporating tiny encoder-decoder network into U-Net. Cross-City Hong et al. (2023) proposes a high-to-low feature fusion network for domain adaptation tasks and achieves state-of-the-art performances in semantic and instance segmentation. A compact version of EfficientNet is proposed to speed up the original approach while maintaining accuracy on ImageNet, efficient for mobile deployment Van Thanh et al. (2025). Although CNNs models achieve high efficiency, the networks have weak receptive fields and modeling abilities Mehta and Rastegari (2022, 2023). Convolution tends to capture local interactions between tokens, and the models require a large number of stacked convolutional blocks across stages to learn global interactions.

2.2. Transformers-based approaches

Recently, a deep learning technique based on transformers Dosovitskiy et al. (2021), Touvron et al. (2021) has been proposed. The vision transformer approach (ViT) presents a new approach to modeling spatial image tokens and has been used in industrial applications Hong et al. (2024), Jiang et al. (2023), Yao et al. (2024). Typically, the image is first divided into non-overlapping patches, which are then linearly embedded and combined with positional embeddings before being processed through a transformer encoder. DeiT Touvron et al. (2021) has achieved high performance on ImageNet and other datasets. At the same time, it may only require a small fraction of training data compared to previous methods. This is thanks to improved training processes and distillation procedures, making it a better choice than ViT when dealing with smaller datasets. Although ViTs create an alternative way of modeling global spatial interactions, they have high computation costs and ignore local modeling between neighborhood tokens. Many works are proposed to reduce the costs of ViTs such as downsampled key and value features Wang et al. (2021, 2022a), local attention Jiang et al. (2023), Liu et al. (2021b), and self-attentions applied on low-resolution features Mehta and Rastegari (2022, 2023). Swin Transformer Jiang et al. (2023), Liu et al. (2021b) introduced window-based partitioning to vision transformers, achieving linear complexity by performing multi-head self-attention within non-overlapping local windows. To model cross-window dependencies, Swin employs shifted windows that cyclically shift the window partition and use masked attention to handle displaced windows. This design reduces complexity from $O(N^2)$ to $O(w^2N)$, making transformers practical for dense prediction tasks. While Swin

uses attention-based operations, we employ pure MLP-based spatial mixing. More critically, instead of shifted windows with cyclic permutation, we introduce dilated window sampling that reduces window count from N_w to N_w/d_r , achieving $O(N/d_r)$ complexity. Our work demonstrates that the window-based design pattern pioneered by Swin can be effectively adapted to MLP architectures with even greater efficiency.

Another solution is to combine convolution and self-attention to build hybrid networks Mehta and Rastegari (2022, 2023), Wang et al. (2022a), Yu et al. (2022b). MobileViT Mehta and Rastegari (2022, 2023) exploits self-attention layers at latter stages of MobileNet Sandler et al. (2018) and achieves high efficiency across tasks. MaxViT Tu et al. (2022) presents a relevant comparison as it combines local and global operations through multi-axis attention, consisting of blocked local attention and dilated grid attention. While MaxViT and DMLP share the conceptual framework of integrating local and global receptive fields, they differ fundamentally in mechanism: MaxViT employs self-attention with $O(wN)$ complexity, whereas DMLP uses simple MLP operations with $O(N/d_r)$ linear complexity. This demonstrates that effective local-global spatial integration is achievable without the computational overhead of attention mechanisms, albeit with an accuracy-efficiency trade-off suitable for resource-constrained deployments. In this work, we introduce efficient networks without using self-attention to model global features, and attain better accuracy and efficiency compared to Transformer-based methods.

2.3. Spatial MLP-Mixer

Spatial MLP-Mixer Tolstikhin et al. (2021) has replaced self-attention blocks with spatial MLP-Mixer blocks and achieved similar performances with ViT. Similar to the problem of ViT, Spatial MLP-Mixer still produces quadratic complexity with token lengths. When transferred to the tasks requiring high-resolution inputs, the MLP-based models result in high computational costs. Another research Hong et al. (2018) proposes a linear mixing model to better estimate various spectral variabilities and obtain superior performance. To facilitate cross-patch interactions, S^2 -MLP Yu et al. (2022a) introduces a token-mixing MLP in addition to the channel-mixing MLP. This work investigates using depth-wise convolution Sandler et al. (2018) with a global receptive field and spatial-specific configuration as a token-mixing operation within the MLP-Mixer. In a different approach, a model based on a multi-scale MLP-Mixer Zhang et al. (2022) is applied for image classification. The multi-scale MLP-Mixer incorporates several MLP-Mixers of varying scales. This method segments each 2D image into a sequence of patches based on the patch size, using these patches as input. In the field of image matching, Mixer-WMLP architecture Shen et al. (2024) performs spatial mixing on partitioned regions and requires special cyclic shifting across regions. The cyclic shifting is not supported by deep learning frameworks and creates additional memory access and latency. ResMLP Touvron et al. (2022) replaces normalization layers with Affine transformation and keeps the MLP-Mixer module as spatial mixing. gMLP Liu et al. (2021a) extends a MLP-Mixer module with spatial gating unit. AS-MLP Lian et al. (2022) introduces axial shifts along height and width dimensions separately, capturing local dependencies through shift operations. While flexible with input sizes, AS-MLP's axial shifts provide only local receptive fields. CycleMLP Chen et al. (2023) employs cyclic FC operators that sample points in a cyclical pattern across the spatial dimension. Although CycleMLP achieves linear complexity and handles variable sizes, its cyclical sampling may lose important visual cues, particularly for small objects in dense prediction tasks. Our DMLP differs by using dilated window partitioning, which maintains spatial structure while achieving superior $O(N/d_r)$ complexity. Specifically, we propose a dilated MLP-Mixer that performs spatial mixing on windows and efficiently exchanges information across windows via dilated operations. Our designs can capture both local and global features while other methods only model global spatial mixing on entire tokens Liu et al. (2021a), Tolstikhin et al. (2021), Touvron et al. (2022), multi-scale tokens Zhang

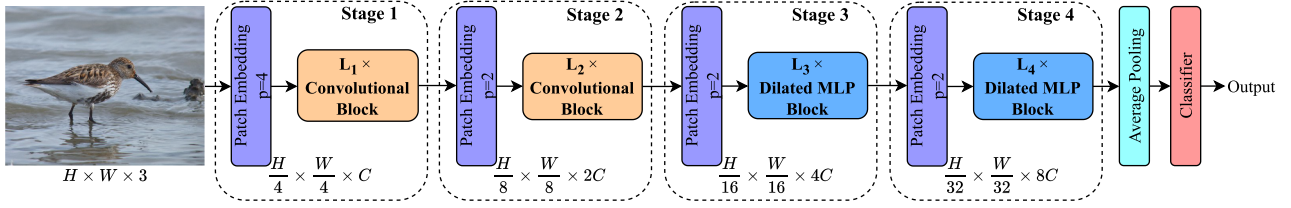


Fig. 1. Overall architecture of the proposed DMLP backbone. DMLP backbone consists of four stages and each stage includes one patch embedding and a stacked number of DMLP blocks. In stage 1 and stage 2, convolutional blocks are introduced to capture local features. Dilated MLP blocks are used in stages 3 and 4 to extract global features. This combination achieves better trade-offs between accuracy and computational costs. H, W, C denote the height, width, and channel dimension of the feature map. L_1, L_2, L_3, L_4 indicates the number of stacked blocks.

et al. (2022) and require convolution inside spatial token mixing module Yu et al. (2022a). Moreover, DMLP-Mixer produces linear complexity with image sequences while methods Liu et al. (2021a), Tolstikhin et al. (2021), Touvron et al. (2022), Yu et al. (2022a), Zhang et al. (2022) still maintain the high cost of the MLP-Mixer module.

Most closely related to our work, RaMLP Lai et al. (2023) proposes Dilated Fully-Connection (DFC) for region-aware mixing. RaMLP's DFC applies dilation to feature partitioning, creating sparse global regions where Spatial FC is performed within each dilated region. This achieves $O(N^2/d_r)$ complexity. In contrast, our DMLP applies dilation to the window partitioning strategy itself, reducing the total number of windows from N_w to N_w/d_r while maintaining dense intra-window connections. This fundamental difference enables DMLP to achieve $O(N/dr)$ complexity - strictly linear rather than RaMLP's quadratic dependence on resolution. While RaMLP achieves higher ImageNet accuracy (83.8% vs our 78.8%), DMLP offers 3.9× computational efficiency and demonstrates strong transfer learning performance on specialized tasks. Swin-MLP Liu et al. (2021b), a window-based MLP method, adapts the shifted window approach to MLP architectures by replacing multi-head self-attention with spatial MLP (group linear layers). Swin-MLP performs spatial mixing within non-overlapping windows and uses shifted windows with cyclic permutation to enable cross-window connections. While Swin-MLP demonstrates the effectiveness of window-based MLP, its shifted window mechanism requires cyclic shift operations and attention masking for boundary handling. Our DMLP differs fundamentally in the cross-window mechanism: instead of shifting the window partition, we introduce dilated window sampling that reduces the number of windows from N_w to N_w/d_r while maintaining dense intra-window connections without requiring cyclic shifting or masking operations.

3. The proposed method

The overall architecture of the proposed DMLP network is shown in Fig. 1. The DMLP network is the hierarchical architecture, consisting of four stages followed by Liu et al. (2021b), Wang et al. (2021, 2022a). Earlier stages tend to capture local features and latter stages tend to extract global features. To achieve this goal, convolutional blocks are used in stages 1 and 2, and Dilated MLP blocks are stacked in stages 3 and 4. Convolutional blocks have locality and linear complexity with the image resolutions, applied in earlier stages to process high-resolution inputs. The Dilated MLP blocks have high flexibility in modeling global receptive fields, used in the latter stages to get better visual representations. This hybrid network can attain good trade-offs between efficiency and accuracy.

Each stage includes one patch embedding and a stacked number of blocks. Patch embedding separates the input feature map into a sequence of patches $\frac{H}{p} \times \frac{W}{p} \times (e * C)$. With four stages, p is a patch size configured to $\{4, 2, 2, 2\}$ and e is a channel expansion configured to $\{1, 2, 4, 8\}$ illustrated in Fig. 1. In practice, patch embedding is implemented by a sequence of convolution and linear projection. Convolution has a kernel size $p \times p$, a stride of p , and the output channel Cp^2 . After convolution, a linear projection is used to control the number of channels from

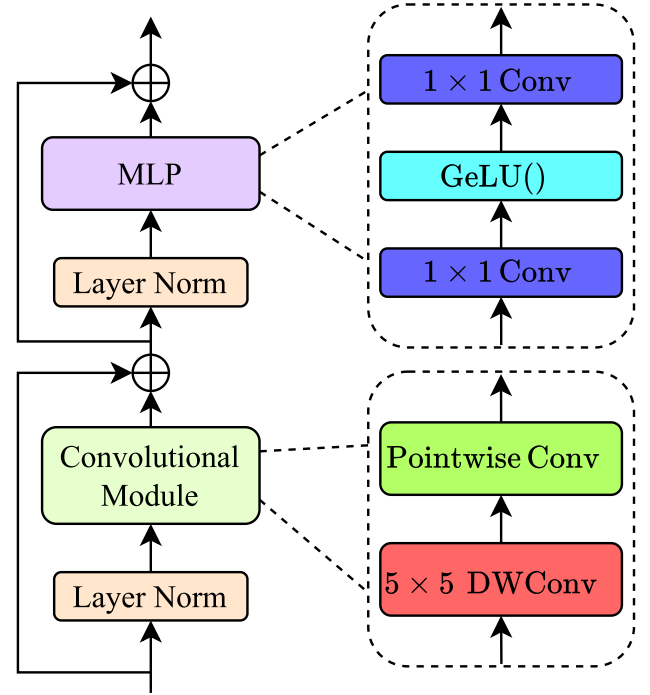


Fig. 2. The detailed structure of the convolutional block.

Cp^2 to $e * C$. Specifically, this projection modifies the output channel dimensions of the Patch Embedding, changing them to $C, 2C, 4C$, and $8C$ across the four stages, respectively. Spatial dimensions are progressively down-sampled while the channel dimensions are increased across four stages. This makes the network deeper and wider. After the four-stage feature extraction, the output feature map from Stage 4 has the spatial size of $\frac{H}{32} \times \frac{W}{32}$ with $8C$ channels. The classifier head is used to process this feature through average pooling, a linear layer, and a softmax function. Specifically, average pooling is employed to aggregate the spatial dimensions of the output feature map from Stage 4, producing a 1D feature vector. This feature vector is then fed into a linear layer that maps the features to the number of target classes (1000 classes in ImageNet-1K image classification). A softmax function is applied to yield the final class probability distribution. The model is trained using cross-entropy loss. In the following, the convolutional and Dilated MLP blocks are discussed in detail.

3.1. Convolutional block

The goal of the convolutional block is to capture local features on high-resolution input. The detailed structure of the convolutional block is shown in Fig. 2. In this structure, $DWConv(\cdot)$ is depthwise convolution and MLP denotes multi-layer perceptron. Given the input feature

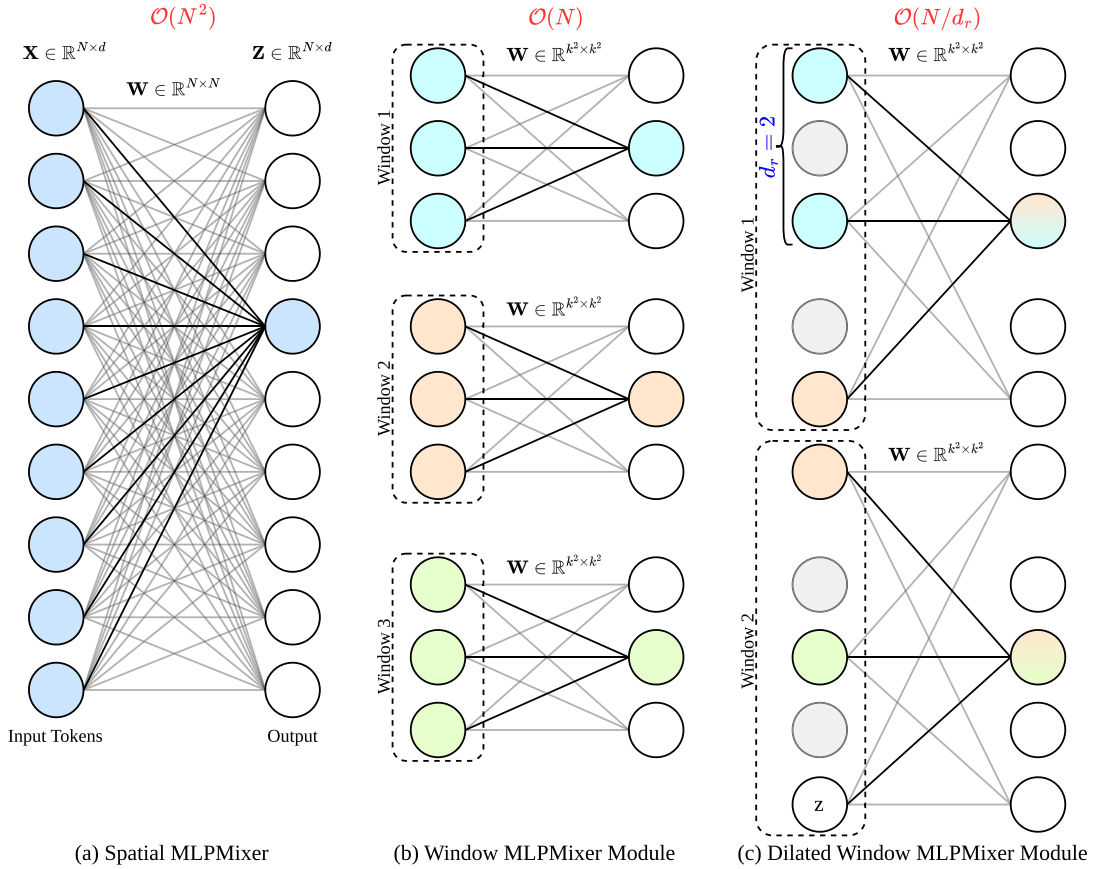


Fig. 3. Comparison between Spatial MLP Mixer Tolstikhin et al. (2021) and the proposed methods. $\mathcal{O}(\cdot)$ denotes complexity of the model. N is the number of tokens and each has d channels, k is window size, d_r indicates dilation rate, and z is the zero-padding values. Each window has three tokens for the simple illustration.

map $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, 5×5 depthwise convolution is adapted to learn local spatial information at low costs, followed by Pointwise Conv to mix information across channel direction. The MLP layer includes two 1×1 convolutions and GeLU() is inserted between them to capture non-linearity from the input. Two layer normalizations are used to normalize the input and stabilize the training process. Similar to meta designs Yu et al. (2022b), two residual connections are utilized to avoid vanishing and exploding gradients. Briefly, these processes are defined as follows:

$$\hat{\mathbf{Y}} = \text{LN}(\text{Convolution}(\mathbf{I})) + \mathbf{I}, \quad (1)$$

$$\mathbf{Y} = \text{LN}(\text{MLP}(\hat{\mathbf{Y}})) + \hat{\mathbf{Y}}, \quad (2)$$

where $\mathbf{Y} \in \mathbb{R}^{H \times W \times C}$ is the output feature map. Convolution($\cdot$) includes two successive operations: DWConv($\cdot$) and PointwiseConv($\cdot$). LN($\cdot$) denotes layer normalization and MLP($\cdot$) indicates multi-layer perceptron.

3.2. Dilated MLP block

The intuitive idea of the Dilated MLP block is to reduce the computational costs of the original Spatial MLP Mixer Tolstikhin et al. (2021) from quadratic to linear complexity and has high flexibility in processing various input sizes instead of fixed dimensions in Spatial MLP Mixer. Fig. 3 shows the comparison between Spatial MLP Mixer and the proposed Window MLP Mixer/Dilated Window MLP Mixer.

Spatial MLP Mixer mixes information across the spatial dimension by performing matrix multiplication between learnable matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ and the input feature $\mathbf{X} \in \mathbb{R}^{N \times d}$. $N = H * W$ is the number of input tokens and d denotes the channel size. As shown in Fig. 3(a), each token is fully connected to all tokens. This process results in global receptive fields. However, Spatial MLP Mixer creates high computational cost and

requires further operations to interpolate the weights. To address these issues, this paper introduces the novel and efficient Window MLP Mixer and Dilated Window MLP Mixer module in Fig. 3(b-c). The Window MLP Mixer captures spatial information inside each window, and the Dilated Window MLP Mixer exchanges information across windows to expand receptive fields and modeling ability. Compared to the quadratic complexity of standard MLP Mixer, our designs achieve linear complexity with image resolutions and do not require interpolation when transferring the model to dense prediction tasks with different input resolutions. We unify Window MLP Mixer module and Dilated Window MLP Mixer module into the DMLP-Mixer module as Window MLP Mixer is a special case of the Dilated Window MLP Mixer module when setting the dilation rate equals 1.

3.2.1. Window MLP Mixer

Given the input token $\mathbf{X} \in \mathbb{R}^{N \times d}$, we reshape and partition $\mathbf{X}$ into a sequence of windows $\mathbf{X}_k \in \mathbb{R}^{N_k \times k^2 \times d}$ where $N_k = \frac{H}{k} * \frac{W}{k}$ is the number of windows across horizontal and vertical axes, k is window size, and d indicates the channel dimension. The features inside each window are learned by the spatial interaction between the window matrix and trainable matrix $\mathbf{W} \in \mathbb{R}^{k^2 \times k^2}$. Technically, the interaction is formulated as:

$$\mathbf{Z}_i = (\mathbf{X}_{k,i}^\top \otimes \mathbf{W})^\top, \quad (3)$$

where $\mathbf{Z}_i \in \mathbb{R}^{k^2 \times d}$ is the output for window i , $\mathbf{X}_{k,i}^\top$ is the transposed input $\mathbf{X}_k$ for i th window, and $\otimes$ indicates matrix multiplication. This computation is processed for all windows to obtain the output with dimension $N_k \times k^2 \times d$. The output is reversed back to its original shape $\mathbf{Z} \in \mathbb{R}^{N \times d}$.

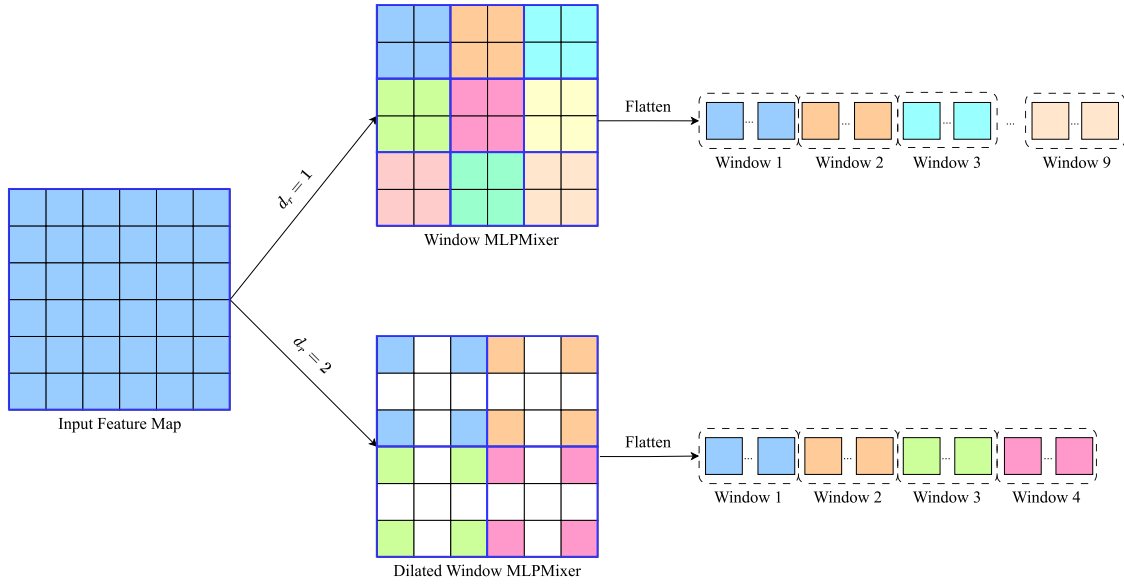


Fig. 4. An illustration of a dilation operation. Colorful tokens are fed into MLP Mixer to capture spatial interaction and white tokens are skipped during interaction. With $d_r = 2$, the number of windows is reduced.

3.2.2. Dilated window MLP Mixer

In the Window MLP Mixer, interactions are computed within non-overlapped windows, limiting receptive fields and having weak modeling ability. To efficiently exchange information across windows, this paper introduces the Dilated Window MLP Mixer module. Fig. 3(c) illustrates the structure of this module. We expand the region for each window with the dilation strategy and keep the window size unchanged via zero-padding. Some skipped tokens are not joined during performing interaction and are kept in the output.

The goal of the dilation is to extract sliding window blocks from the input feature map. Given an input $\mathbf{X} \in \mathbb{R}^{H \times W \times d}$, we unfold $\mathbf{X}$ to the $\mathbf{X}' \in \mathbb{R}^{N_k \times k^2 \times d}$ where N_k is the number of output window, k is window size, and d is the channel number. For each window position $n \in \{1, \dots, N_k\}$, the dilated window is extracted according to:

$$\mathbf{X}'[n, :, :] = \text{flatten}(\mathbf{X}[v_s : v_e : d_r, h_s : h_e : d_r, :]), \quad (4)$$

where the start and end indices are computed as follows:

$$v_s = \lfloor n / N_k^h \rfloor * k - z, \quad (5)$$

$$v_e = v_s + k * d_r, \quad (6)$$

$$h_s = (n \bmod N_k^h) * k - z, \quad (7)$$

$$h_e = h_s + k * d_r, \quad (8)$$

where v_s, v_e are vertical start and end indices. h_s, h_e are the horizontal start and end indices. z is the zero-padding values. d_r is the dilation rate. N_k^h is the number of windows along the horizontal dimension, computed as:

$$N_k^h = \lfloor (W + 2z - d_r(k-1) - 1) / k \rfloor + 1, \quad (9)$$

$$N_k^v = \lfloor (H + 2z - d_r(k-1) - 1) / k \rfloor + 1, \quad (10)$$

$$N_k = N_k^h * N_k^v, \quad (11)$$

where N_k^v is the number of windows along the vertical dimension. Fig. 4 illustrates the mechanism of the dilation operation. After obtaining the dilated feature $\mathbf{X}'$, performing the spatial interaction on $\mathbf{X}'$ can achieve efficient information exchange across windows.

With this design, the number of windows is reduced and the complexity of the model is decreased from $\mathcal{O}(N)$ to $\mathcal{O}(N/d_r)$. As shown in

Fig. 3(c), the receptive field of the model is enlarged while maintaining the efficiency of the Window MLP Mixer. The neighbor windows are connected to update the information.

3.2.3. Dilated MLP block

The Window MLP Mixer module is proposed to learn spatial information inside non-overlapped windows and the Dilated Window MLP Mixer module is introduced to exchange information across neighbor windows. Consecutively stacking two modules into one block can help to learn spatial features effectively. Fig. 5 illustrates the detailed structure of the DMLP block.

DMLP block is built based on the common design Dosovitskiy et al. (2021), Liu et al. (2021b), Wang et al. (2021), consisting of a DMLP-Mixer module (spatial mixing) and a channel mixer module (channel mixing). Two consecutive layers are computed as:

$$\hat{\mathbf{Z}}^l = \text{LN}(\text{W-MLP}(\mathbf{X})) + \mathbf{X} \quad (12)$$

$$\mathbf{Z}^l = \text{LN}(\text{MLP}(\hat{\mathbf{Z}}^l)) + \hat{\mathbf{Z}}^l \quad (13)$$

$$\hat{\mathbf{Z}}^{l+1} = \text{LN}(\text{DW-MLP}(\mathbf{Z}^l)) + \mathbf{Z}^l \quad (14)$$

$$\mathbf{Z}^{l+1} = \text{LN}(\text{MLP}(\hat{\mathbf{Z}}^{l+1})) + \hat{\mathbf{Z}}^{l+1} \quad (15)$$

where l indicates the l^{th} layer. W-MLP($\cdot$) is Window MLP Mixer module and DW-MLP($\cdot$) is Dilated Window MLP Mixer module.

Similar to the MLP-Mixers Liu et al. (2021a), Shen et al. (2024), Tolstikhin et al. (2021), Touvron et al. (2022), Zhang et al. (2022), we do not supplement absolute positional embedding Dosovitskiy et al. (2021), Touvron et al. (2021) or adaptive positional embedding Wang et al. (2022a) into the image features to learn positional relations between image tokens. MLP-Mixers and our DMLP-Mixer still have a weak ability in modeling positional relations. In future works, we will inject absolute or adaptive positional embedding with the image features to improve the performance.

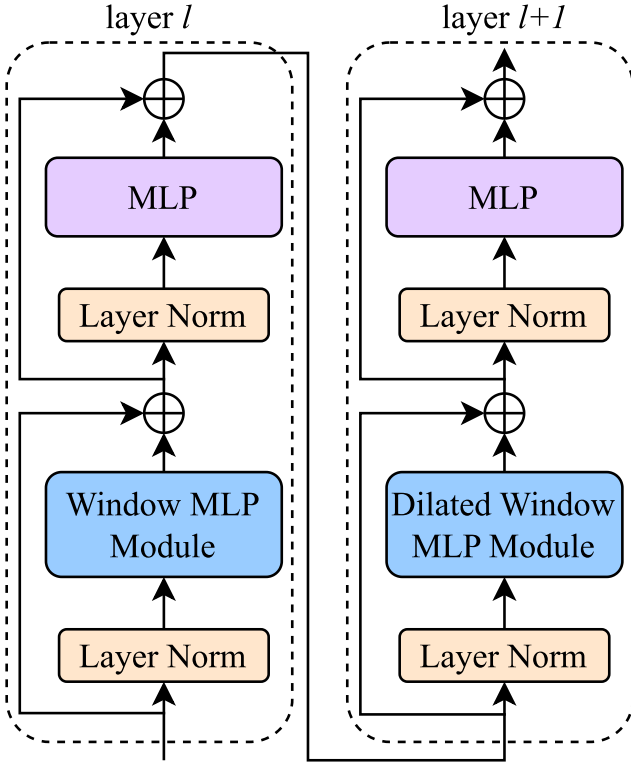


Fig. 5. Two consecutive layers of the DMLP network.

Table 2
Model Configuration of DMLP variants.

Model	#Channels	#Blocks	MLP exp.	#Params(M)
DMLP-XTiny	[24, 48, 96, 192]	[2, 2, 3, 3]	[4, 4, 4, 4]	6.5
DMLP-Tiny	[32, 64, 128, 256]	[2, 2, 3, 3]	[4, 4, 4, 4]	10.3
DMLP-Small	[48, 96, 192, 384]	[2, 2, 4, 3]	[4, 4, 4, 4]	22.1

3.3. Model complexity and configuration

The proposed DMLP-Mixer, including WindowMLP-Mixer (W-MLP) and Dilated WindowMLP-Mixer (DW-MLP), achieves higher efficiency than Spatial MLP-Mixer [Tolstikhin et al. \(2021\)](#). Spatial mixing is performed within non-overlapped windows by evenly partitioning the image into window regions. The complexity of the original MLP-Mixer and our method is computed as follows:

$$\mathcal{O}(\text{Spatial-MLP-Mixer}) = N^2, \quad (16)$$

$$\begin{aligned} \mathcal{O}(\text{DMLP-Mixer}) &= \mathcal{O}(\text{W-MLP}) + \mathcal{O}(\text{DW-MLP}) \\ &= k^2 N + k^2 N / d_r \\ &\approx N + N / d_r. \end{aligned} \quad (17)$$

As the window size k is constant and much smaller than the number of image tokens N , the complexity of DMLP-Mixer is approximated as shown in the above equation. During training, we set $k = 7$ for all blocks. Therefore, the proposed method has much smaller costs than the original Spatial MLP-Mixer. For example, in stage 1 of the image classification model (spatial size 56×56 , channel dimension 32), DMLP-Mixer results in 4.9 MFLOPs while Spatial-MLP-Mixer produces 314.7 MFLOPs. It verifies the efficiency of the proposed method.

Model Configuration. Based on the obtained block, DMLP variants are constructed by controlling the number of blocks and channels across four stages. The general architecture of the DMLP is discussed in [Fig. 1](#). [Table 2](#) shows model configurations of the DMLP variants. #Channels denote the number of channels across stages, controlled by patch em-

bedding layers. #Blocks is the number of blocks across stages. MLP exp. is the MLP expansion ratio and is kept unchanged across blocks.

4. Experiments

The DMLP backbone is evaluated and compared on two situations of the general purpose and the specific purpose [Hoang et al. \(2024\)](#). The specific purpose of car damage recognition in applications related to car insurance monitoring. The car damage dataset is highly challenging. Datasets consist of small damage areas that are difficult to detect with the human eye, resulting in low accuracy using AI applications. In this experiment, the proposed method is compared with recent methods on both criteria of accuracy and processing speed. Speed comparisons between efficient methods are performed on low-cost devices such as CPU and GPU with limited memories.

4.1. Experiment on the general-purpose dataset

DMLP models are validated on ImageNet-1K image classification, MS-COCO object detection/instance segmentation, and ADE-20K semantic segmentation. Firstly, DMLP methods are trained on the ImageNet-1K training set and the performance is reported on the ImageNet-1K validation set. Secondly, the pre-trained models on the ImageNet-1K dataset are transferred to dense prediction tasks to verify the efficient and general-purpose DMLP backbones.

4.1.1. Image classification

Setting: ImageNet-1K dataset consists of 1.2M training and 50K validation images, used to train and evaluate DMLP models. Following training recipes [Liu et al. \(2021b\)](#), [Tolstikhin et al. \(2021\)](#), [Touvron et al. \(2021\)](#), [Wang et al. \(2021\)](#), the proposed networks are trained for 300 epochs on four GPU RTX 3090s, with a batch size of 2,048. AdamW is used as an optimizer with a learning rate of 4×10^{-3} scheduled by cosine learning, and a weight decay of 0.03. Rand Augment, CutMix, MixUp, and label smoothing are adopted to train the model, as standard data augmentation [Tolstikhin et al. \(2021\)](#), [Touvron et al. \(2021\)](#). The input images are resized to 224×224 . Throughput is measured on the NVIDIA GTX 1070 and CPU Intel(R) Core(TM) i5-6600 CPU @ 3.30GHz with a batch size of 256 and 16.

Results: In the problem of applying proposed solutions to image classification tasks, we trained the classification model on the well-known dataset of ImageNet1k and evaluated the proposed model on the respective validation set. The proposed method was compared with some method groups, such as (1) MLP based methods, (2) Transformer based methods, and (3) CNN based methods. The details of comparison results are demonstrated in [Table 3](#).

The experimental results illustrate that three architectures of the proposed method, DMLP-XTiny, DMLP-Tiny, and DMLP-Small, achieve accuracy, comparable to the state-of-the-art methods. However, our proposed method has significantly fewer parameters and lower GFLOPs, which compared to other methods. In the case of MLP-based methods, although ResMLP-S24 and gMLP-S models reach slightly higher accuracy (about 0.7%) but that methods are larger numbers of parameters and noticeable higher GFLOPs at 6.0 and 4.5 compared to the DMLP-Small method at 2.0, respectively. Similarly, transformer-based methods also have a significant number of parameters. Specifically, the DeiT-S model achieves slight accuracy compared to DMLP-Small but it is 4.6 GFLOPs compared to 2.0 GFLOPs respectively. Furthermore, CNN-based methods achieve lower accuracy but are also equivalent in terms of the number of parameters and GFLOPs. We also report throughput comparisons on GPU and CPU between methods. As a result, the proposed DMLP achieves better efficiency than other methods. We also provide the speed comparison between our methods and recent methods using CPU ONNX runtime. Our DMLP achieves better trade-offs between accuracy and speed compared to Transformer-like methods and CNN-like methods. Typically, DMLP-XTiny surpasses efficient MobileViTv2-0.5 by

Table 3

Results of the proposed DMLP and other methods for image classification with ImageNet-1K.

Model	Res.	#Params	GFLOPs	Top1-Acc(%)	Throughput(imgs/s)			
					GPU	CPU	ONNX Runtime	
MLP-like Methods								
Mixer-B/16 Tolstikhin et al. (2021)	224 ²	59.0M	12.7	76.4	167	3	8	
Mixer-L/16 Tolstikhin et al. (2021)	224 ²	207.0M	44.8	71.8	50	1	2	
ResMLP-S12 Touvron et al. (2022)	224 ²	15.0M	3.0	76.6	618	13	21	
ResMLP-S24 Touvron et al. (2022)	224 ²	30.0M	6.0	79.4	314	7	11	
gMLP-Ti Liu et al. (2021a)	224 ²	6.0M	1.4	72.3	598	22	18	
gMLP-S Liu et al. (2021a)	224 ²	20.0M	4.5	79.6	263	7	14	
SwinMLP-T/C24 Liu et al. (2021b)	256 ²	20.0M	4.0	79.4	257	8	16	
SwinMLP-T/C12 Liu et al. (2021b)	256 ²	21.0M	4.0	79.6	258	5	14	
SwinMLP-B Liu et al. (2021b)	224 ²	61.0M	10.4	81.3	129	3	7	
AS-MLP-T Lian et al. (2022)	224 ²	28.0M	4.4	81.3	-	-	-	
CycleMLP-B1 Chen et al. (2023)	224 ²	15.0M	2.1	79.1	267	5	-	
CycleMLP-B2 Chen et al. (2023)	224 ²	27.0M	3.9	81.6	163	3	-	
Hire-MLP-Tiny Guo et al. (2022)	224 ²	18.0M	2.1	79.7	410	6	19	
Hire-MLP-Small Guo et al. (2022)	224 ²	33.0M	4.2	82.1	215	4	8	
Ra-MLP-T Lai et al. (2023)	224 ²	25.0M	4.2	82.9	76	3	6	
Transformer-like Methods								
ViT-B/16 Dosovitskiy et al. (2021)	384 ²	86.0M	55.4	77.9	34	1	1	
ViT-L/16 Dosovitskiy et al. (2021)	384 ²	307.0M	190.7	76.5	51	1	OOM	
DeiT-T Touvron et al. (2021)	224 ²	5.0M	1.3	72.2	657	28	41	
DeiT-S Touvron et al. (2021)	224 ²	22.0M	4.6	79.8	369	9	13	
PVTv1-T Wang et al. (2021)	224 ²	13.2M	1.9	75.1	450	14	23	
PVTv2-B0 Wang et al. (2022a)	224 ²	3.4M	0.6	70.5	666	25	35	
PoolFormer-S12 Yu et al. (2022b)	224 ²	11.9M	1.8	77.2	570	12	25	
CNN-like Methods								
MobileViTv1-XXS Mehta and Rastegari (2022)	256 ²	1.3M	0.4	69.0	548	19	32	
MobileViTv1-XS Mehta and Rastegari (2022)	256 ²	2.3M	1.0	74.8	252	8	15	
MobileViTv1-S Mehta and Rastegari (2022)	256 ²	5.6M	2.0	78.4	203	5	11	
MobileViTv2-0.5 Mehta and Rastegari (2023)	256 ²	1.4M	0.3	70.2	524	19	38	
MobileViTv2-0.75 Mehta and Rastegari (2023)	256 ²	2.9M	1.0	75.5	326	9	23	
MobileViTv2-1.0 Mehta and Rastegari (2023)	256 ²	4.9M	1.9	78.1	247	6	17	
ResNet-18 He et al. (2016)	224 ²	11.7M	1.8	69.8	1227	18	57	
ResNet-50 He et al. (2016)	224 ²	25.6M	4.1	76.1	444	7	34	
ResNet-101 He et al. (2016)	224 ²	44.7M	7.9	77.4	256	4	16	
Our proposed methods								
DMLP-XTiny	224 ²	6.5M	0.5	74.4	673	38	56	
DMLP-Tiny	224 ²	10.3M	0.8	76.8	525	26	24	
DMLP-Small	224 ²	22.1M	2.0	78.8	303	13	22	

*OOM: Out of Memory

Table 4

Top-1 accuracy on transfer learning datasets.

Model	#params	GFLOPs	CIFAR-10	CIFAR-100	Cars-196
ResNet-50 He et al. (2016)	26	4.1	-	-	90.0
ViT-B/16 Dosovitskiy et al. (2021)	86	55.4	98.1	87.1	-
ViT-L/16 Dosovitskiy et al. (2021)	307	190.7	97.9	86.4	-
DeiT-B/16 Touvron et al. (2021)	86	17.5	99.1	90.8	92.1
ResMLP-12 Touvron et al. (2022)	15	3.0	98.1	87.0	84.6
ResMLP-24 Touvron et al. (2022)	30	6.0	98.7	89.5	89.5
DMLP-Small	22	2.0	98.5	89.1	92.4

4.2% Top-1 accuracy while running 1.4 times faster than MobileViTv2-0.5 on the CPU ONNX runtime.

Transfer Learning. We have transferred DMLP-Small on common benchmark datasets, CIFAR-10 [Krizhevsky et al. \(2009\)](#), CIFAR-100 [Krizhevsky et al. \(2009\)](#), and Stanford Cars [Krause et al. \(2013\)](#). We follow the training recipe of methods [Dosovitskiy et al. \(2021\)](#), [Tolstikhin et al. \(2021\)](#), [Touvron et al. \(2021\)](#) and the models pre-trained on ImageNet-1K are fine-tuned on three datasets. Table 4 reports Top-1 accuracy on the new datasets. As a result, DMLP-Small attains competitive performances with MLP-based methods (ResMLP-24) while using significantly fewer computational resources (2.0 vs 6.0 GFLOPs). Notably, DMLP-Small achieves the highest accuracy on Cars-196 (92.4%) among all methods, surpassing the Transformer-based method DeiT-B/16 (92.1%) while requiring 3.9× fewer parameters (22M vs 86M) and

Table 5

Ablation study on window sizes using DMLP-XTiny.

Window Size	#params(M)	GFLOPs	Top-1 Acc (%)
3	6.275	0.475	73.9
5	6.328	0.403	74.1
7	6.500	0.488	74.4
9	6.700	0.472	74.5

8.75× fewer GFLOPs (2.0 vs 17.5). It verifies that the proposed method has generalization ability on datasets.

Ablation Study. Table 5 reports the performance when ablating the window sizes on ImageNet-1K validation set. As a result, when window sizes are increased, performance is improved. In this configuration, we

Table 6
Investigations of positions of DMLP Block across four stages.

Model	Positions of DMLP Block				Top-1 Acc. (%)	#params (M)	GFLOPs
	Stage 1	Stage 2	Stage 3	Stage 4			
Model 1	✓	✓	✓	✓	75.3	6.788	0.782
Model 2	✗	✓	✓	✓	74.9	6.684	0.603
Model 3	✗	✗	✓	✓	74.4	6.500	0.488
Model 4	✗	✗	✗	✓	72.8	5.746	0.369
Model 5	✗	✗	✗	✗	69.7	3.085	0.222

Table 7
Effectiveness of the Dilated Window MLP Module using DMLP-XTiny.

Module		Top-1 Acc. (%)	#params (M)	GFLOPs
Window MLP	Dilated Window MLP			
✓		73.1	4.519	0.330
	✓	73.6	4.519	0.281
✓	✓	74.4	6.500	0.488

set the window size to 7 as this setting achieves a better tradeoff between the number of parameters and Top-1 accuracy. Note that the computational cost depends on the number of windows and window size. Configuring window size to be smaller results in a large number of windows, and otherwise.

Table 6 reports the performances of the proposed Dilated MLP block versus the convolutional block on ImageNet-1K dataset. ✓ indicates that only DMLP blocks are used at this stage and ✗ denotes that only convolutional blocks are utilized at this stage. All the models are trained and evaluated under the same experimental setting.

- Model 5: The convolutional blocks are positioned at four stages, capturing the locality of features (i.e., inductive bias). The main drawback of convolution is the model has weak receptive field and modeling ability. In a hierarchical structure, earlier stages (stage 1 and stage 2) capture local features and later stages (stage 3 and stage 4) tend to extract global features. Hence, stacking convolutional blocks at later stages can lead to weak global modeling ability.
- Model 4: We replace convolutional blocks at stage 4 with DMLP blocks. As a result, the proposed DMLP improves the model 5 by 3.1% Top-1 accuracy. This is due to that DMLP models both local and global features. Adopting DMLP blocks at later stages can achieve better feature modeling.
- Model 3: When using DMLP blocks at stage 3&4 and convolutional blocks at stage 1&2, the model achieves 74.4% Top-1 accuracy that greater than model 4 by 1.6%. It verifies the modeling ability of the proposed method.
- Model 1&2: Using DMLP block at earlier stages (stage 1 and stage 2) still achieves consistent improvements on ImageNet-1K dataset. Because the DMLP Block performs spatial mixing on non-overlapped windows and exchanges cross-window information to expand receptive fields, one block can capture local and global features. Otherwise, convolutional blocks (model 5) only extract local features and the model requires a large number of stacked blocks to expand receptive fields.

The computational cost of DMLP module depends on the number of windows and window sizes. When using DMLP modules at earlier stages with high-resolution inputs, it creates larger number of windows and hence, increases GFLOPs. As shown in Table 2, Model 1 surpasses Model 3 by 0.9% Top-1 accuracy with 0.294 GFLOPs increasing. To achieve the balance between computational costs and accuracy improvements, we use Model 3 configuration for all experiments.

The goal of the Dilated Window MLP module is to exchange information across windows efficiently, capturing global features from the input. To verify the effectiveness of the Dilated Window MLP module,

Table 8
Ablation study on dilation rate using DMLP-XTiny.

Dilation rate d_r	#params (M)	GFLOPs	Top-1 Acc. (%)
1	4.519	0.330	73.1
2	6.500	0.488	74.4
3	6.500	0.439	74.3
4	6.500	0.416	74.3

Table 9
Results of the proposed DMLP for object detection using SSD-Lite.

Backbone	#params(M)	GFLOPs	mAP
MobileNetv1 Sandler et al. (2018)	5.1	1.3	22.2
MobileNetv2 Sandler et al. (2018)	4.3	0.8	22.1
MobileViTv1-XXS Mehta and Rastegari (2022)	1.7	0.9	19.9
MobileViTv2-0.5 Mehta and Rastegari (2023)	2.0	0.9	21.2
ResNet-50 He et al. (2016)	26.6	8.8	25.2
DMLP-XTiny	6.2	0.8	21.6
DMLP-Tiny	9.9	1.5	22.1
DMLP-Small	21.7	3.5	25.6

Table 10
Results of the proposed DMLP for object detection using RetinaNet.

Backbone	#params(M)	GFLOPs	AP	AP ₅₀	AP ₇₅
ResNet-18 He et al. (2016)	21	189	31.7	49.6	33.4
ResNet-50 He et al. (2016)	38	250	36.3	55.3	38.6
PVT-T Wang et al. (2021)	23	183	36.7	56.6	38.8
PoolFormer-S12 Yu et al. (2022b)	22	207	36.2	56.2	38.2
DMLP-Tiny	18	168	35.0	55.1	37.0
DMLP-Small	30	192	36.6	56.9	38.7

we only used Window MLP module in the model DMLP-XTiny. Table 7 reports the three cases: DMLP-XTiny model with only Window MLP, DMLP-XTiny model with only Dilated Window MLP, and DMLP-XTiny with both Window MLP and Dilated Window MLP. The results show that both modules are effective when used independently. The DMLP-XTiny with only Dilated Window MLP achieves 73.6% Top-1 accuracy, outperforming the Window MLP only configuration (73.1%) by 0.5% while requiring fewer GFLOPs (0.281 vs. 0.330). This verifies that the proposed dilation mechanism effectively captures global context with superior efficiency. Furthermore, combining both modules yields the highest accuracy of 74.4%, indicating that the Window MLP and Dilated Window MLP modules provide complementary features: local spatial interactions and cross-window dependencies.

Table 8 shows an ablation study on the dilation rate of Dilated Window MLP module using DMLP-XTiny model. When $d_r = 1$, we only use Window MLP module, as Dilated Window MLP with $d_r = 1$ is functionally equivalent to the Window MLP module. Other configurations employ both Window MLP and Dilated Window MLP with varying dilation rates. Increasing d_r can capture information exchange across windows. As reported in Table 8, the increase of dilation rate boosts the performance of the model. Among all configurations, DMLP-XTiny with $d_r = 2$ achieves the highest accuracy while maintaining acceptable computa-

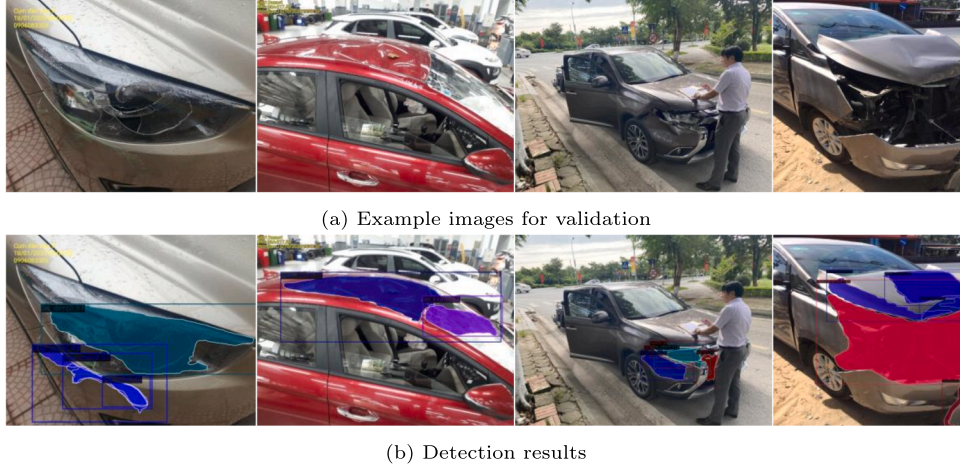


Fig. 6. Examples of Car damage dataset and detection results.

Table 11

Results of the proposed DMLP for object detection using Mask R-CNN.

Backbone	#params(M)	GFLOPs	AP ^{box}	AP ^{mask}	AP ^{mask} ₅₀	AP ^{mask} ₇₅
ResNet-18 He et al. (2016)	31	207	34.0	31.2	51.0	32.7
ResNet-50 He et al. (2016)	44	260	38.0	34.4	55.1	36.7
PVT-T Wang et al. (2021)	33	208	36.7	35.1	56.7	36.3
PoolFormer-S12 Yu et al. (2022b)	31	-	37.3	34.6	55.8	36.9
DMLP-Tiny	29	186	36.2	33.9	55.3	36.0
DMLP-Small	40	210	38.5	35.8	57.2	37.9

Table 12

Results of DMLP for semantic segmentation using SemanticFPN.

Method	#Params	GFLOPs	mIoU
ResNet-18 He et al. (2016)	15.6M	32.2	32.9
ResNet-50 He et al. (2016)	28.5M	45.6	36.7
PVT-T Wang et al. (2021)	17.0M	33.2	35.7
DMLP-Tiny	12.7M	25.2	34.3
DMLP-Small	24.5M	31.1	36.0

Table 13

Details of Car damage dataset.

Damage categories	train	val
Lost parts	2,370	448
Torn	4,546	963
Dents	4,709	972
Scratch	12,267	2,380
Broken glass	1,824	397
Punctured	2,006	417
Broken lights	2,324	458
Total	30,046	6,035

tional costs. Accordingly, we adopt $d_r = 2$ as the default setting for all experiments.

4.1.2. Object detection and instance segmentation

Setting: After finishing experiments on ImageNet-1K, we transfer trained weights for object detection and instance segmentation tasks using three detectors: SSD-Lite [Liu et al. \(2016\)](#), RetinaNet [Lin et al. \(2017\)](#), and Mask R-CNN [He et al. \(2017\)](#). The original ResNet-50 [He et al. \(2016\)](#) backbone in RetinaNet and Mask R-CNN, VGG [Simonyan and Zisserman \(2014\)](#) in SSD are replaced with efficient methods.

Table 14

Results of Car damage recognition using RetinaNet.

Method	#params	GFLOPs	AP ^{box}	AP ⁵⁰	AP ⁷⁵	FPS
ResNet-50	36.2M	207.0	23.1	41.7	21.9	29.4
DMLP-Tiny	16.6M	135.3	24.0	44.0	22.3	31.1
DMLP-Small	28.6M	159.8	24.7	44.3	23.6	29.8

For SSD-Lite [Liu et al. \(2016\)](#), we follow training settings of lightweight CNNs [Sandler et al. \(2018\)](#) and MobileViT family [Mehta and Rastegari \(2022, 2023\)](#). Specifically, the models are trained for 120 epochs with a batch size of 144. The optimizer is AdamW with a learning rate of 10^{-3} , and a weight decay of 0.05. The input images are resized to 320×320 .

For RetinaNet [Lin et al. \(2017\)](#) and Mask R-CNN [He et al. \(2017\)](#), the training settings are kept the same as recent methods [Liu et al. \(2021b\)](#), [Wang et al. \(2021\)](#). The learning schedule is 12 epochs with a batch size of 16. The optimizer is AdamW with a learning rate of 10^{-3} and a weight decay of 0.05. The input images are resized to $1,333 \times 800$.

Results: When transferring DMLP models to downstream tasks, we attain consistent improvements across detectors. Typically, with detector SSD-Lite [Liu et al. \(2016\)](#) in Table 9, DMLP-XTiny outperforms mobile vision transformers MobileViTv1-XXS [Mehta and Rastegari \(2022\)](#) by 1.7% AP, and MobileViTv2-0.5 by 0.9% AP. DMLP-Small surpasses ResNet-50 by 0.4% AP while saving half of GFLOPs.

Table 10 reports performance when using RetinaNet [Lin et al. \(2017\)](#). DMLP-Tiny achieves 35%AP that is greater than ResNet-18 [He et al. \(2016\)](#) by 3.2% with lower parameters and GFLOPs. Meanwhile, our proposed approach for DMLP-Small outperforms the baseline ResNet-50 by 0.3%AP with saving 58 GFLOPs and lower parameters. Additionally, our proposed approach also outperforms PoolFormer-S12 model by 0.4%AP with saving 15 GFLOPs. The interesting of the important issue in our method is that the DMLP-Small model has more parameters compared to the PoolFormer-S12 model. However, the proposed

Table 15
Results of Car damage recognition using Mask R-CNN.

Backbone	#params	GFLOPs	AP ^{box}	AP ^{box} ₅₀	AP ^{box} ₇₅	AP ^{mask}	AP ^{mask} ₅₀	AP ^{mask} ₇₅	FPS
ResNet-50	43.4M	258.3	23.8	43.2	23.1	19.8	38.3	18.0	12.1
DMLP-Tiny	28.2M	183.9	23.9	44.5	22.4	20.5	39.2	18.8	14.3
DMLP-Small	40.0M	208.7	24.5	45.2	23.7	21.0	39.8	19.6	12.5

method is based on the dilated MLP-Mixer technique, which reduces computational complexity to $O(N/d_r)$, thereby achieving better efficiency in GFLOPs. Finally, DMLP-Small achieves superior performance compared to PVT-T (1.8% higher AP^{box} and 0.7% higher AP^{mask}), despite having more parameters (40M vs 33M). Importantly, both methods maintain comparable computational efficiency (210 vs 208 GFLOPs), verifying the effectiveness of MLP-like methods on various visual tasks. Table 11 shows the results when integrating the proposed backbone and other backbones into Mask R-CNN He et al. (2017). We achieve similar improvements compared to other backbones. For instance, DMLP-Small gets 38.5 AP^{box} and 35.8 AP^{mask} which is better than the baseline ResNet-50 by 0.5% AP^{box} and 1.4% AP^{mask} with lower parameters and GFLOPs. It clarifies the generalization ability of the proposed method.

4.1.3. Semantic segmentation

Setting: The DMLP models are validated on ADE-20K using Semantic FPN. Similar to Liu et al. (2021b), Wang et al. (2021, 2022a), the models are trained for 80K iterations with a batch size of 16. AdamW is used as an optimizer with a learning rate of 10^{-3} , and a weight decay of 0.05. The input images are resized to 512×512 .

Results: Table 12 reports the comparison between the proposed method and previous methods. DMLP-Tiny achieves 34.3% mIoU that surpasses ResNet-18 He et al. (2016) by 1.4% with the lower number of parameters and GFLOPs. DMLP-Small outperforms the Vision Transformer method, PVT-T Wang et al. (2021), by 0.3% mIoU with similar costs. The results show that our MLP-based method achieves superior performances compared to CNN-based and Transformer-based methods.

4.2. Car damage recognition

The proposed DMLP method was evaluated with the specialized dataset of automobile insurance problem for car damage recognition Hoang et al. (2024). The Car Damage dataset consists of 13,945 images and 36,081 objects within 7 categories. The training set contains 11,621 images with 30,046 objects, and the validation set contains 2,324 images with 6,035 objects, see also Table 13 for more details. The examples of dataset are illustrated in Fig. 6a. This dataset is challenging for abnormal objects detecting and recognizing. Some damaged areas are very small or very similar to reflection light, stains on the car surfaces such as paint sheen or mud. As a result, the state-of-the-art backbones- based semantic segmentation, anomaly detection methods tend to reach low accuracy. The Car Damage dataset poses a challenge for practical applications in intelligent monitoring systems.

Setting: Firstly, DMLP methods are trained on the ImageNet-1K training set. Secondly, the pretrained models on the ImageNet-1K dataset are transferred to train on MS-COCO for object detection. Finally, three backbones are transferred learning on Car damage dataset for two detectors RetinaNet Lin et al. (2017), and Mask R-CNN He et al. (2017). The SSD method was not evaluated due to it is not suitable for small object detection and the results are very low precision in the specialized application of car damage recognition.

Results: The results in Table 14 show that RetinaNet detector Lin et al. (2017) using ResNet50 backbone is the largest number of parameters and FLOPs meanwhile the lowest AP values. The detector using DMLP-Tiny reaches AP that is 0.9% higher and 2.3% higher in AP50 and 0.6% higher in AP75. DMLP-Small performs better than ResNet50 with

a 1.6% improvement in AP^{box} , 2.6% in AP_{50} , and 1.7% in AP_{75} . Experimental results in Table 15 also demonstrate similar results for damage recognition using Mask R-CNN He et al. (2017). The visualization of detection results using Mask R-CNN is demonstrated in Fig. 6b. The proposed backbones for the feature extraction task deliver superior accuracy compared to previous approaches. Meanwhile, the backbones have fewer parameters and lower computational costs than recent methods.

5. Conclusion

This paper introduces a new MLP method, dubbed Dilated MLP-Mixers (DMLP), addressing the issues of previous MLP methods. First, DMLP reduces the cost of spatial MLP-Mixer from quadratic to linear complexity via performing token mixing on windows. With this proposed design, DMLP enjoys higher efficiency while capturing both local and global receptive fields. Second, our method has high flexibility when transferring the model to dense prediction tasks requiring high-resolution inputs. Conventional Spatial MLP-Mixers need further operations to interpolate MLP weights, leading to lower performances compared to CNN-based and Transformer-based methods. This is solved by our DMLP method while achieving similar or even better results with CNN-based and Transformer-based backbones. Extensive experiments demonstrate the efficiency and effectiveness of the proposed method across dense prediction tasks. Unlike common research trends that try to improve accuracy by using larger or more complex models, this paper focuses on efficiency. Our approach increases accuracy while reducing computational costs and the number of parameters. This research direction could help develop intelligent systems for industries with limited computing power and storage.

CRedit authorship contribution statement

Van-Dung Hoang: Writing – review & editing, Validation, Supervision, Methodology, Data curation; **Xuan-Thuy Vo:** Writing – original draft, Validation, Methodology, Conceptualization; **Kang-Hyun Jo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by Ho Chi Minh City University of Technology and Engineering, Vietnam, under grant No. T2026-10.

References

- Chen, S., Xie, E., Ge, C., Chen, R., Liang, D., & Luo, P. (2023). Cyclemlp: A mlp-like architecture for dense visual predictions. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(12), 14284–14300.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International conference on learning representations*.
- Guo, J., Tang, Y., Han, K., Chen, X., Wu, H., Xu, C., Xu, C., & Wang, Y. (2022). Hire-mlp: Vision mlp via hierarchical rearrangement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 826–836).

- He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961–2969).
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).
- Hoang, V.-D., Huynh, N. T., Tran, N., Le, K., Le, T.-M.-C., Selamat, A., & Nguyen, H. D. (2024). Powering AI-driven car damage identification based on veHIDE dataset. *Journal of Information and Telecommunication*, (pp. 1–19).
- Hong, D., Yokoya, N., Chanussot, J., & Zhu, X. X. (2018). An augmented linear mixing model to address spectral variability for hyperspectral unmixing. *IEEE Transactions on Image Processing*, 28(4), 1923–1938.
- Hong, D., Zhang, B., Li, H., Li, Y., Yao, J., Li, C., Werner, M., Chanussot, J., Zipf, A., & Zhu, X. X. (2023). Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks. *Remote Sensing of Environment*, 299, 113856.
- Hong, D., Zhang, B., Li, X., Li, Y., Li, C., Yao, J., Yokoya, N., Li, H., Ghamisi, P., Jia, X. et al. (2024). SpectralGPT: Spectral remote sensing foundation model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, .
- Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. arXiv preprint arXiv:1704.04861.
- Hu, H., Yu, C., Zhou, Q., Guan, Q., & Feng, H. (2024). Hdconv: Heterogeneous kernel-based dilated convolutions. *Neural Networks*, 179, 106568.
- Jiang, J., Zhu, J., Bilal, M., Cui, Y., Kumar, N., Dou, R., Su, F., & Xu, X. (2023). Masked swin transformer unet for industrial anomaly detection. *IEEE Transactions on Industrial Informatics*, 19(2), 2200–2209.
- Krause, J., Stark, M., Deng, J., & Fei-Fei, L. (2013). 3D object representations for fine-grained categorization. In *Proceedings of the IEEE international conference on computer vision workshops* (pp. 554–561).
- Krizhevsky, A., Hinton, G. et al. (2009). Learning multiple layers of features from tiny images, .
- Lai, S., Du, X., Guo, J., Zhang, K., & Elkind, E. (2023). RaMLP: Vision MLP via region-aware mixing. In *Ijcai* (pp. 999–1007).
- Lian, D., Yu, Z., Sun, X., & Gao, S. (2022). As-mlp: An axial shifted mlp architecture for vision. In *International conference on learning representations (ICLR)*.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision* (pp. 2980–2988).
- Liu, H., Dai, Z., So, D., & Le, Q. V. (2021a). Pay attention to mlps. *Advances in Neural Information Processing Systems*, 34, 9204–9215.
- Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., & Berg, A. C. (2016). Ssd: Single shot multibox detector. In *Computer vision—ECCV 2016: 14th european conference, amsterdam, the netherlands, october 11–14, 2016, proceedings, part i 14* (pp. 21–37). Springer.
- Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021b). Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 10012–10022).
- Mehta, S., & Rastegari, M. (2022). Mobilevit: Light-weight, general-purpose, and mobile-friendly vision transformer. In *International conference on learning representations*.
- Mehta, S., & Rastegari, M. (2023). Separable self-attention for mobile vision transformers. *Transactions on Machine Learning Research*, .
- Peng, Y., Liu, H., Kong, W., Nie, F., Lu, B.-L., & Cichocki, A. (2023). Joint EEG feature transfer and semisupervised cross-subject emotion recognition. *IEEE Transactions on Industrial Informatics*, 19(7), 8104–8115.
- Pham, T.-A., & Hoang, V.-D. (2024). Chest x-ray image classification using transfer learning and hyperparameter customization for lung disease diagnosis. *Journal of Information and Telecommunication*, (pp. 1–15).
- Pham, T.-C., Luong, C.-M., Hoang, V.-D., & Doucet, A. (2021). Ai outperformed every dermatologist in dermoscopic melanoma diagnosis, using an optimized deep-cnn architecture with custom mini-batch logic and loss function. *Scientific Reports*, 11(1), 17485.
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4510–4520).
- Shen, Z., Kong, B., & Dong, X. (2024). Maim: A mixer mlp architecture for image matching. *The Visual Computer*, 40(3), 1327–1337.
- Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556, .
- Tolstikhin, I. O., Houlsby, N., Kolesnikov, A., Beyer, L., Zhai, X., Unterthiner, T., Yung, J., Steiner, A., Keysers, D., Uszkoreit, J. et al. (2021). Mlp-mixer: An all-mlp architecture for vision. *Advances in Neural Information Processing Systems*, 34, 24261–24272.
- Touvron, H., Bojanowski, P., Caron, M., Cord, M., El-Nouby, A., Grave, E., Izacard, G., Joulin, A., Synnaeve, G., Verbeek, J. et al. (2022). Resmlp: Feedforward networks for image classification with data-efficient training. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4), 5314–5321.
- Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., & Jégou, H. (2021). Training data-efficient image transformers & distillation through attention. In *International conference on machine learning* (pp. 10347–10357). PMLR.
- Tu, Z., Talebi, H., Zhang, H., Yang, F., Milanfar, P., Bovik, A., & Li, Y. (2022). Maxvit: Multi-axis vision transformer. In *European conference on computer vision* (pp. 459–479). Springer.
- Van Thanh, H., Quang, N. D., Phuong, T. M., Jo, K.-H., & Hoang, V.-D. (2025). A compact version of efficientnet for skin disease diagnosis application. *Neurocomputing*, 620, 129166.
- Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 568–578).
- Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2022a). Pvt v2: Improved baselines with pyramid vision transformer. *Computational Visual Media*, 8(3), 415–424.
- Wang, Z., Jiang, W., Zhu, Y. M., Yuan, L., Song, Y., & Liu, W. (2022b). Dynamixer: A vision mlp architecture with dynamic mixing. In *International conference on machine learning* (pp. 22691–22701). PMLR.
- Wu, X., Hong, D., & Chanussot, J. (2022). Uiu-net: U-net in u-net for infrared small object detection. *IEEE Transactions on Image Processing*, 32, 364–376.
- Xu, Y., Feng, K., Yan, X., Sheng, X., Sun, B., Liu, Z., & Yan, R. (2024). Cross-modal fusion convolutional neural networks with online soft-label training strategy for mechanical fault diagnosis. *IEEE Transactions on Industrial Informatics*, 20(1), 73–84.
- Yao, H., Cao, Y., Luo, W., Zhang, W., Yu, W., & Shen, W. (2024). Prior normality prompt transformer for multi-class industrial image anomaly detection. *IEEE Transactions on Industrial Informatics*, .
- Yu, T., Li, X., Cai, Y., Sun, M., & Li, P. (2022a). S2-Mlp: Spatial-shift mlp architecture for vision. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision* (pp. 297–306).
- Yu, W., Luo, M., Zhou, P., Si, C., Zhou, Y., Wang, X., Feng, J., & Yan, S. (2022b). Metaformer is actually what you need for vision. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10819–10829).
- Zhang, H., Dong, Z., Li, B., & He, S. (2022). Multi-scale MLP-mixer for image classification. *Knowledge-Based Systems*, 258, 109792.
- Zhang, Y., Pan, H., Li, M., Wang, A., Zhou, Y., & Ren, H. (2024). Spatial-wise dynamic distillation for MLP-like efficient visual fault detection of freight trains. *IEEE Transactions on Industrial Electronics*, .
- Zhou, W., Zheng, F., Zhao, Y., Pang, Y., & Yi, J. (2024). Msdcnn: A multiscale dilated convolution neural network for fine-grained 3d shape classification. *Neural Networks*, 172, 106141.
- Zhuang, Y., Zheng, Z., Zhang, Y., Lyu, L., Jia, X., & Lyu, C. (2024). Dimensional transformation mixer for ultra-high-definition industrial camera dehazing. *IEEE Transactions on Industrial Informatics*, .